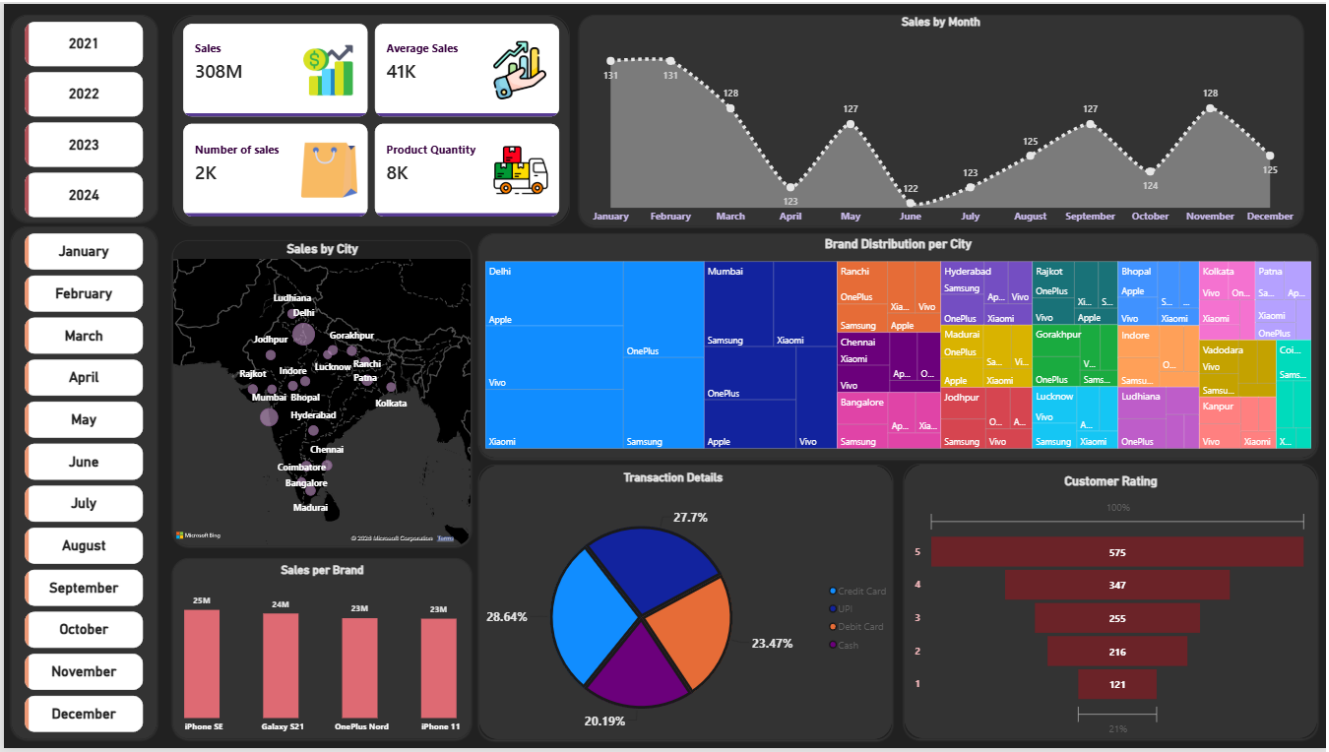
PRODUCT REQUIREMENTS DOCUMENT

Mobile Sales Performance & Customer Insights Dashboard

A requirements baseline for the interactive business intelligence dashboard shown in the supplied reference image.



Contents

- 1. Executive summary
- 2. Product overview and objectives
- 3. Users and use cases
- 4. Scope and exclusions
- 5. Dashboard layout and requirements
- 6. Filters, interactions and behaviour
- 7. Data requirements and business definitions
- 8. Non-functional requirements
- 9. Acceptance criteria and test scenarios

1. Executive summary

The Mobile Sales Performance & Customer Insights Dashboard is an interactive business intelligence product intended to provide a consolidated view of sales performance, product demand, geographic distribution, payment behaviour and customer ratings. The reference design presents four headline KPI cards, year and month selectors, and six analytical visualisations. The dashboard should help business users identify performance patterns, compare markets and products, and investigate changes by time period.

This PRD translates the visible design into implementation-ready requirements. The screenshot does not provide the underlying dataset, calculation logic, refresh schedule or complete business definitions. Accordingly, numerical values visible in the image are treated as illustrative display values, not as validated targets or source-of-truth results.

2. Product overview and objectives

2.1 Product vision

Provide a clear, filterable, single-screen view of mobile-device sales and customer indicators so that business users can move from high-level performance monitoring to city-, brand-, month- and transaction-level analysis.

2.2 Business objectives

- Monitor total sales value, average sales value, transaction count and product quantity in one place.
- Understand monthly sales movement and identify high- and low-performing periods.
- Compare sales contribution across cities and mobile brands.
- Understand the distribution of transaction payment methods.
- Review customer rating distribution and identify potential experience issues.
- Allow users to filter the complete report by year and month without navigating between multiple pages.

2.3 Success measures

Measure	Proposed success criterion	Measurement method
KPI consistency	All KPI cards reconcile to the approved source data and documented definitions.	Reconciliation test against source totals.
Filter reliability	Every supported slicer updates all applicable visuals consistently.	Filter interaction test matrix.
Usability	A business user can identify the selected reporting period and interpret each visual without developer assistance.	Stakeholder walkthrough / usability review.
Performance	Initial report load and common filter interactions meet the agreed performance target.	Timed test using representative data volume.
Data freshness	Displayed data is refreshed within the agreed refresh service level.	Refresh timestamp and pipeline monitoring.

3. Users and use cases

User group	Primary need	Typical activity
Sales manager	Track overall sales and monthly performance.	Review KPIs, identify monthly trends and compare cities.
Regional / city manager	Understand local market performance.	Filter by period and compare city-level brand distribution.
Product / category manager	Evaluate brand and product demand.	Compare sales by brand and product quantity.
Business analyst	Investigate patterns and validate reported figures.	Cross-filter visuals, inspect breakdowns and reconcile measures.
Senior leadership	Obtain a concise performance overview.	Review headline KPIs and major trends.

3.1 Core user journeys

1. Select a reporting year and month, then review the refreshed KPI cards and charts.
2. Identify a month with unusually high or low sales and compare it with other months.
3. Select or focus on a city to understand the brands contributing to that city's sales.
4. Compare brand-level sales and product demand for the selected reporting period.
5. Review payment method mix and customer rating distribution to understand transaction and experience patterns.

4. Scope and exclusions

4.1 In scope

- A single-page interactive dashboard based on the supplied visual structure.
- Year selector with the years 2021, 2022, 2023 and 2024, subject to data availability.
- Month selector with January through December.
- Four KPI cards: Sales, Average Sales, Number of Sales and Product Quantity.
- Monthly sales trend, sales by city map, brand distribution by city treemap, sales per brand, transaction payment-method mix and customer rating distribution.
- Consistent cross-filtering and/or cross-highlighting across visuals where the underlying data relationship supports it.
- Clear titles, units, labels, tooltips and selected-filter context.

4.2 Out of scope unless separately approved

- Order entry, billing, inventory management or payment processing.
- Customer-level personally identifiable information or individual customer profiles.
- Predictive forecasting, recommendations or automated anomaly detection.
- Write-back, approval workflows, alerts and scheduled email distribution.
- Drill-through to invoice/order records unless the required detail data and security model are provided.
- Additional pages, mobile-app deployment or public embedding.

5. Dashboard layout and requirements

The implementation should follow the reference image's dark, high-contrast layout while maintaining readable typography, consistent spacing and accessible colour contrast. Exact pixel-level replication is not required unless separately specified.

ID	Component	Requirement	Priority
UI-01	Year selector	Display selectable years 2021–2024. The selected year must be visually obvious and must filter all applicable report content.	Must
UI-02	Month selector	Display all 12 calendar months. Support a single selected month; multi-select behaviour must be agreed before build.	Must

KPI-01	Sales	Display total sales value for the active filter context, formatted using an agreed currency and compact notation where appropriate.	Must
KPI-02	Average Sales	Display average sales value using the approved business definition (proposed: total sales value divided by transaction count).	Must
KPI-03	Number of Sales	Display the count of qualifying sales transactions/orders, based on an agreed unique transaction key.	Must
KPI-04	Product Quantity	Display total units sold for the active filter context.	Must
VIS-01	Sales by Month	Show monthly sales trend with month labels and data labels or tooltips. Months with no sales should be represented consistently.	Must
VIS-02	Sales by City	Show geographic sales distribution by city using map markers sized or coloured by an agreed sales measure. Provide city name and sales value in tooltip.	Must
VIS-03	Brand Distribution per City	Show a treemap grouped by city and subdivided by brand. Rectangle size must represent a documented measure, such as sales value or quantity.	Must
VIS-04	Sales per Brand	Show sales value by brand in a comparable bar/column chart. Sort order and treatment of ties must be defined.	Must
VIS-05	Transaction Details	Show the percentage distribution of transactions by payment method (e.g., Credit Card, UPI, Debit Card, Cash). Percentages should total 100% subject to rounding and exclusions.	Must
VIS-06	Customer Rating	Show the count of ratings for each rating value (1–5), with a clear scale and count labels. Rating coverage should be defined.	Must
UX-01	Visual consistency	Use consistent number formatting, title casing, colour mapping and tooltip conventions throughout the report.	Should
UX-02	Filter context	Show selected year/month clearly and provide a reset-to-default option.	Must

5.1 Visual-specific behaviour

Visual	Expected behaviour / tooltip content
KPI cards	Show measure name, formatted value and (if approved) comparison to prior period. Do not show a comparison until its baseline and calculation are defined.
Monthly sales trend	Tooltip: month, sales value and optionally transaction count. Selecting a month should filter or highlight other visuals.
City map	Tooltip: city, sales value, transaction count and quantity if available. Map should not imply exact location precision beyond city-level data.
City-brand treemap	Tooltip: city, brand and the measure used for rectangle size. Clicking a segment should filter/highlight the other visuals where supported.
Sales per brand	Tooltip: brand, sales value and optionally quantity sold. Use a consistent brand colour if the same brand appears elsewhere.
Payment method chart	Legend must identify each payment method. Tooltip: payment method, transaction count and percentage.
Customer rating chart	Tooltip: rating value, response count and percentage of valid ratings.

6. Filters, interactions and behaviour

6.1 Global filters

Filter	Input behaviour	Expected effect
Year	Single-select buttons for 2021–2024. Default selection must be agreed; recommended default is latest complete year with data.	Filters all KPIs and visuals to the selected year.
Month	Single-select buttons for January–December. Default may be All Months if supported by the report design; otherwise define a default month.	Filters all applicable KPIs and visuals to the selected month.
Reset	A visible reset control or bookmark/action.	Restores the agreed default year and month and clears visual selections.

6.2 Cross-filtering

- Selecting a data point in a chart should cross-filter or cross-highlight other visuals according to the BI platform's interaction settings.

- A city selection should narrow brand distribution and other city-sensitive measures where data relationships permit.
- A brand selection should update the relevant sales, quantity and city breakdowns.
- A payment method or rating selection should filter other visuals only if the business approves that analytical behaviour and the data model supports it.
- Users must be able to clear a visual selection without losing the global year/month filter.

6.3 Empty and exceptional states

- If a filter combination returns no records, display a clear 'No data for the selected filters' message rather than misleading zeros or a blank visual.
- If a city cannot be geocoded, include it in a clearly identified unmapped category or exception list; do not silently drop its sales.
- If rating or payment method is missing, report the missing/unknown category or disclose its exclusion from the denominator.
- If a measure is not applicable, display 'N/A' rather than a fabricated value.
- Long city or brand names must remain readable through wrapping, truncation with tooltip, or an equivalent approach.

7. Data requirements and business definitions

The source system and schema have not been supplied. The following is a proposed minimum data contract and must be validated with the data owner before implementation.

7.1 Proposed source fields

Field	Description	Requirement / validation
Transaction ID	Unique identifier for a sale/order.	Required; uniqueness and duplicate handling must be defined.
Transaction Date	Date/time of the sale.	Required; valid date, agreed timezone and reporting calendar.
Sales Amount	Monetary value attributed to the transaction or line item.	Required; currency, tax, discount, refund and cancellation treatment must be defined.
Product ID / Product Name	Identifier and display name for the item sold.	Required for product-level analysis.
Brand	Brand associated with the product.	Required for brand visuals; standardise spelling and null handling.
Quantity	Number of units sold.	Required; define treatment of returns and negative quantities.
City	City associated with the sale/customer/store.	Required for city analysis; standardised city dimension and mapping.
Payment Method	Payment channel used for the transaction.	Required for payment mix; controlled category list.
Customer Rating	Customer rating, expected scale 1–5.	Optional if not collected for every transaction; define valid values and response grain.

7.2 Proposed measure definitions

Measure	Proposed calculation	Decision required
Total Sales	SUM of eligible sales amount for the current filter context.	Confirm gross vs net sales, currency, taxes, discounts, returns and cancellations.
Average Sales	Total Sales ÷ Number of Sales.	Confirm whether average is per transaction, order, invoice or line item.
Number of Sales	Distinct count of the approved transaction/order identifier.	Confirm what constitutes a sale and whether cancelled/refunded orders are excluded.
Product Quantity	SUM of eligible quantity.	Confirm returns, bundles, exchanges and unit-of-measure rules.
Monthly Sales	Total Sales grouped by calendar month and selected year.	Confirm calendar vs fiscal month and treatment of partial months.
Sales per Brand	Total Sales grouped by brand.	Confirm whether the chart should instead represent quantity.
Payment Method %	Transactions for a payment method ÷ eligible	Confirm split/partial payments and unknown

	transactions with known payment method.	payment method handling.
Customer Rating Count	Count of valid rating responses grouped by rating value.	Confirm one rating per transaction/customer and treatment of invalid/missing values.

7.3 Data quality rules

- Transaction identifiers must be non-null and conform to the agreed uniqueness rule.
- Sales amount and quantity must be numeric and use documented sign conventions.
- Dates outside the supported reporting range must be flagged for review.
- Brand, city and payment method values must map to approved dimension values; unmapped values must be visible to the data owner.
- Rating values must fall within the approved 1–5 scale; invalid values must be excluded or separately reported.
- The same transaction must not be double-counted because of joins between transaction, product and rating tables.

8. Non-functional requirements

ID	Category	Requirement	Priority
NFR-01	Performance	Target initial load under 5 seconds and common filter response under 3 seconds for the agreed representative dataset and deployment environment.	Should
NFR-02	Accuracy	All displayed measures must reconcile to the approved source and documented business definitions.	Must
NFR-03	Availability	Report availability must meet the organisation's agreed BI service target and planned maintenance policy.	Should
NFR-04	Security	Access must follow role-based permissions and the organisation's data access policies.	Must
NFR-05	Privacy	Do not expose personally identifiable customer data in this dashboard unless separately approved and justified.	Must
NFR-06	Usability	Labels, titles, legends and tooltips must be understandable to non-technical business users.	Must
NFR-07	Accessibility	Maintain readable text size and sufficient contrast; do not rely on colour alone to convey meaning.	Should
NFR-08	Responsive layout	Dashboard should remain usable at the agreed target screen sizes; smaller screens may use a simplified layout if necessary.	Should
NFR-09	Maintainability	Measures, data transformations and relationships must be documented and use consistent naming conventions.	Should
NFR-10	Refresh	Refresh frequency, completion window, failure notification and displayed last-refresh timestamp must be agreed.	Must

9. Acceptance criteria and test scenarios

The dashboard is ready for stakeholder acceptance when all Must requirements pass and any accepted deviations are documented.

Test ID	Scenario	Expected result
AT-01	Open the dashboard with the agreed default state.	All visuals render; default year/month are visible; no unexplained errors.
AT-02	Select each available year.	All applicable visuals and KPIs update to the selected year.
AT-03	Select each month.	All applicable visuals update; monthly trend and KPI values follow the agreed month-filter behaviour.
AT-04	Select a city on the map.	City-sensitive visuals respond according to the approved interaction design.
AT-05	Select a brand segment in the treemap.	Brand-sensitive visuals respond; selected brand/city context is clear.
AT-06	Compare KPI values with an independently calculated source extract.	Sales, average sales, transaction count and quantity reconcile within agreed rounding tolerance.
AT-07	Validate payment method distribution.	Percentages reconcile to 100% within rounding tolerance, using the approved denominator.
AT-08	Validate customer rating distribution.	Counts match valid source ratings; missing/invalid ratings are handled as documented.
AT-09	Apply a filter combination with no matching records.	A clear no-data state is displayed; no misleading values are shown.
AT-10	Use reset control after multiple selections.	All filters and visual selections return to the documented default state.
AT-11	Test representative performance and screen sizes.	Performance and layout meet agreed NFR targets.
AT-12	Test user permissions.	Users can access only the data and report features authorised for their role.